1. $y'' + 7y = 10e^{2t}$

$$L\{y''\} + 7L\{y\} = 10\frac{1}{s-2}$$

$$s^2 L\{y\} - 3y(0) - y'(0) + 7L\{y\}$$

$$(s^2+7)\,L\{y\} - 3 = \frac{10}{s-2}$$

$$L\{y\} = \frac{10}{(s-2)(s^2+7)} + \frac{3}{s^2+7}$$

$$\frac{10}{(s-2)(s^2+7)} = \frac{A}{s-2} + \frac{Bs+C}{s^2+7}$$

$$= As^2 + 7A + (Bs+C)(s-2)$$

$$= s^2(A) + 7A + Bs^2 - 2Bs + Cs - 2C$$

$$= s^2(A+B) + s(C-2B) + 7A - 2C$$

$A+B=0$

$C-2B=0$ $\quad$ $C+2A=0$ $\quad$ $2C+4A=0$

$7A-2C=10$ $\quad$ $7A-2C=10$ $\quad$ $7A-2C=10$

$\boxed{B=\frac{-10}{11}}$ $\quad$ $C+\frac{20}{11}=0$ $\quad$ $11A=10$

$\boxed{A=10/11}$

$\boxed{C=-20/11}$

$$10/11\,\frac{1}{s-2} + \frac{(-10/11)s}{s^2+7} - \frac{20}{11}\left(\frac{1}{s^2+7}\right)$$

$$10/11\,e^{2t} - \frac{10}{11}\cos\sqrt{7}t - \frac{20}{11\sqrt{7}}\sin\sqrt{7}t + \frac{3}{\sqrt{7}}\left(\frac{1}{s^2+7}\right)$$

$$= \frac{10}{11}e^{2t} - \frac{10}{11}\cos\sqrt{7}t + \frac{13}{11\sqrt{7}}\sin\sqrt{7}t$$

2. $y'' - 3y' - 10y = 2$ $\qquad y(0) = 1$

$y'(0) =$

$$L\{y''\} - 3L\{y'\} - 10L\{y\} = 2/s$$

$$L\{y''\} = s^2 L\{y\} - s\,y(0) - y'(0)$$
$$L\{y'\} = s L\{y\} - y(0)$$

$$s^2 L\{y\} - s - 2 - 3[s L\{y\} - 1] - 10L\{y\} = 2/s$$

$$s^2 L\{y\} - s - 2 - 3s L\{y\} - 3 - 10L\{y\} = 2/s$$

$$L\{y\}[s^2 - 3s - 10] - s(1) - 5 = 2/s$$

$$L\{y\}[s^2 - 3s - 10] - s(\quad) = 2/s + 5$$

$$L\{y\}[\quad] = 2/s + 5 + s$$

$$L\{y\} = \frac{2}{s(s-5)(s+2)} + \frac{5}{(s-5)(s+2)} + \frac{s}{(s-5)(s+2)}$$

$$2 = A(s-5)(s+2) + Bs(s+2) + Cs(s-5)$$

$$A = -1/5 \quad B = 2/35 \quad C = 1/3$$

$$-1/5 \; \frac{1}{s} + 2/35 \; \frac{1}{s-5} + \frac{1}{3} \; \frac{1}{s+2} + \frac{1}{(s-5)(s+2)} + \frac{s}{(s-5)(s+2)}$$

$$\frac{-1}{5}\frac{1}{s} + \frac{2}{35}\frac{1}{(s-5)} + \frac{1}{3}\frac{1}{(s+2)} + \frac{1}{7}\frac{1}{(s+2)} - \frac{1}{7}\frac{1}{(s-5)}$$

$$\frac{s}{(s-5)(s+2)} = \frac{A}{(s-5)} + \frac{B}{(s+2)}$$

$$As + 2A + Bs - 5B = s$$

$$A + B = 1$$

$$2A - 5B = 0$$

$$A = \frac{5B}{2} \qquad \frac{5B}{2} + B = 1 \qquad B = 2/7$$

$$7B = 1$$

$$\boxed{A = 5/7}$$

$5/7\, e^{5t} + 2/7\, e^{2t}$

$-1/5 + 2/35\, e^{5t} + 1/3\, e^{-2t} + 1/3\, e^{-2t} - 1/7\, e^{5t}$

$- 5/7\, e^{5t} + 2/7\, e^{-2t}$

$-1/5 + e^{5t}\left(2/35 - 1/7 - 5/7\right) + e^{-2t}\left(1/3 + 1/7 + 2/7\right)$

$-1/5 - 4/5\, (e^{5t}) + e^{-2t}\left(16/21\right)$

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3. $y'' - y' - 2y = e^{2t}$

$L\{y''\} - L\{y'\} - 2L\{y\} = \frac{1}{s-2}$

$s^2 L\{y\} - s \times 0 - 1 - [sL\{y\} - 1] - 2L\{y\} = \frac{1}{s-2}$

$s^2 L\{y\} - \cancel{1} - sL\{y\} + \cancel{1} - 2L\{y\} = \frac{1}{s-2}$

$L\{y\}[s^2 - s - 2] = \frac{1}{s-2} + 1$

$L\{y\} = \frac{1}{(s-2)(s-2)(s+1)} + \frac{1}{(s-2)(s+1)}$

$= \frac{1}{(s-2)^2(s+1)} = \frac{A}{(s-2)^2} + \frac{B}{(s-2)} + \frac{C}{s+1}$

$1 = \frac{A(s+1) + B(s-2)(s+1) + C(s-2)^2}{(s-2)^2(s-1)}$

$As + A + s^2B - sB - 2B + Cs^2 + 4C - 4sC$

$s^2(B+C) + s(A - B - 4C) + A - 2B + 4C$

$B + C = 0$

$A - B - 4C = 0$

$A - 2B + 4C = 1$

$A = 1/3 \quad C = 1/9$

$B = -1/9$

$\frac{1}{(s-2)(s+1)} = 1/3 \frac{1}{(s-2)} - \frac{1}{3(s+1)}$

$1/3\, e^{2t} - 1/3\, e^{-t}$

ANS $= 1/3\, te^{2t} + 2/9\, e^{2t} - 4/9\, e^{-t}$

$1/3 \frac{1}{(s-2)^2} - 1/9 \frac{1}{(s-2)} + 1/9 \frac{1}{(s+1)}$

$1/3\, te^{2t} - 1/9\, e^{2t} + 1/9\, e^{-t}$

B. 4. $y'' - 2y' + 2y = e^{-t}$ $\quad y(0) = 0 \quad y'(0) = 1$

$$L\{y''\} - 2L\{y'\} + 2L\{y\} = \frac{1}{s+1}$$

$$s^2 L\{y\} - s(0) - 1 - 2[sL\{y\} - 0] + 2L\{y\} = \frac{1}{s+1}$$

$$s^2 L\{y\} - 1 - 2sL\{y\} + 2L\{y\} = \frac{1}{s+1}$$

$$(s^2 - 2s + 2)L\{y\} = \frac{1}{s+1} + 1$$

$$L\{y\} = \frac{1}{(s+1)(s^2-2s+2)} + \frac{1}{(s^2-2s+2)}$$

$$\frac{A}{(s+1)} + \frac{Bs+C}{s^2-2s+2} = \frac{1}{(s+1)(s^2-2s+2)}$$

$$As^2 - 2As + 2s + Bs^2 + Bs + Cs + C = 1$$

$$s^2(A+B) + s(-2A + 2 + B + C) + C = 1$$

$$\boxed{C = 1}$$

$$\boxed{A + B = 0} \qquad A = 1$$

$$-2A + 2 + B + C = 0 \qquad B = -1$$

$$-2A + B + C = -2$$

$$\frac{1}{s+1} + \frac{-s+1}{s^2+2-2s}$$

$$e^{-t} - \frac{s-1}{s^2+1-2s+1}$$

$$e^{-t} - \frac{(s-1)}{(s-1)^2+1}$$

$$e^{-t} - e^{t}\cos t + e^{t}\sin t$$

$$\frac{1}{(s-1)^2+1}$$

$$e^{t}\sin t$$

5. $y'' + y = \cos 2t \quad y(0) = 2, \quad y'(0) = 1$

$$L\{y''\} + L\{y\} = \frac{s}{s^2+4}$$

$$s^2 L\{y\} - s2 - 1 + L\{y\} = \frac{s}{s^2+4}$$

$$L\{y\}(s^2+1) - 2s - 1 = \frac{s}{s^2+4}$$

$$L\{y\}(s^2+1) = \frac{s}{s^2+4} + 1 + 2s$$

$$L\{y\} = \frac{s}{(s^2+4)(s^2+1)} + \frac{1}{s^2+1} + \frac{2s}{s^2+1}$$

$$\frac{As+B}{s^2+4} + \frac{Cs+D}{s^2+1} = \frac{s}{(s^2+4)(s^2+1)}$$

$$(As+B)(s^2+1) + (Cs+D)(s^2+4) = s$$

$$As^3 + As + Bs^2 + B + Cs^3 + Cs + Ds^2 + 4D = s$$

$$s^3(A+C) + s^2(B+D) + s(A+4C) + B + 4D = s$$

$A + C = 0$

$B + D = 0 \qquad A = -1/3 \qquad C = 1/3$

$A + 4C = 1 \qquad B = 0 \qquad D = 0$

$B + 4D = 0$

$$-1/3 \frac{s}{s^2+4} + 1/3 \frac{s}{s^2+1} + \frac{1}{s^2+1} + 2\frac{s}{s^2+1}$$

$1/3 + 2$

$\frac{1+6}{3}$

$$-1/3 \cos 2t + 1/3 \cos t + \sin t - 2\cos t$$

$+7/3$

$$-1/3 \cos 2t + \sin t + 7/3 \cos t.$$

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